

$$\mathcal{L} = \mathcal{L}_{\text{ERP}} + w \mathcal{L}_{\text{design}} + 0.1 \mathcal{L}_{\text{pad}}$$

$$\mathcal{L}_{\text{ERP}} = \text{mean} (\hat{y} - y)^2$$

$$\mathcal{L}_{\text{design}} = \text{Huber}(\hat{a}, a), \quad \mathcal{L}_{\text{pad}} = \text{Huber}(\hat{z}, 0)$$

$$w = \min (1, (e + 1)/10)$$

a : resonator design (m, f_t, x, y) of each resonator; $\hat{a}$: design recovered from the true ERP by the inverse

z : extra padded values of the design (zero going forward); $\hat{z}$: their recovered values

y : true ERP; $\hat{y}$: predicted ERP; e : current epoch; E : total number of epochs